UNIT-4

IMAGE SEGMENTATION

Introduction

Image segmentation is a fundamental task in computer vision that involves partitioning an image into multiple meaningful regions or segments. Unlike object detection or classification, which focus on identifying objects within an image as a whole, segmentation aims to identify and delineate individual objects or regions within the image.

The goal of image segmentation is to simplify and/or change the representation of an image into something that is more meaningful and easier to analyze. It plays a crucial role in various applications such as medical image analysis, autonomous driving, satellite image analysis, and more.

There are several types of image segmentation techniques:

1. **Thresholding**: This technique involves selecting a threshold value and assigning all pixels in the image above or below this threshold to different segments.
2. **Region-based segmentation**: In this approach, pixels with similar properties such as color, intensity, texture, or motion are grouped together to form regions.
3. **Edge-based segmentation**: Edge detection algorithms are used to identify boundaries between different objects or regions in an image.
4. **Clustering-based segmentation**: Techniques like k-means clustering group similar pixels together based on some feature space.
5. **Semantic segmentation**: This technique involves assigning each pixel in an image to a specific class label, such as "car", "building", "sky", etc.
6. **Instance segmentation**: Similar to semantic segmentation, but it also distinguishes between different instances of the same class, providing a pixel-level mask for each individual object.

Edge detection

Edge detection is a fundamental step in image segmentation, particularly in methods that rely on edge-based segmentation. The primary goal of edge detection is to identify the boundaries or edges between different objects or regions in an image. These boundaries typically correspond to significant changes in intensity or color within the image.

Common Edge Detection Techniques

1. Sobel Operator

The Sobel operator computes the gradient of the image intensity at each pixel, giving the direction of the largest possible increase from light to dark and the rate of change in that direction.

2. Prewitt Operator

Similar to the Sobel operator, the Prewitt operator is used to compute the gradient of the image. It uses a different kernel for the convolution.

3. Canny Edge Detector

The Canny edge detector is a multi-stage algorithm that is more sophisticated and widely used for edge detection.

Steps in Canny Edge Detection

1. **Noise Reduction**: Apply a Gaussian filter to smooth the image and reduce noise.
2. **Gradient Calculation**: Compute the intensity gradients of the image.
3. **Non-Maximum Suppression**: Thin out the edges to produce thin lines by suppressing non-maximum gradient values.
4. **Double Thresholding**: Apply a threshold to classify strong, weak, and non-relevant pixels.
5. **Edge Tracking by Hysteresis**: Finalize edge detection by suppressing all edges that are not connected to strong edges.

Edge Linking Via Hough Transform

Edge linking via the Hough transform is a powerful technique in image segmentation used to detect and link edges into geometric shapes, such as lines, circles, and other parametric shapes. The Hough transform is particularly effective in identifying straight lines within an image, even in the presence of noise or partial occlusion. Here's an overview of how this technique works and how it can be applied in edge linking for image segmentation.

Key Concepts

1. **Edge Detection**: The process of identifying significant changes in intensity in an image. Common edge detection methods include the Sobel, Canny, and Prewitt operators.
2. **Hough Transform**: A feature extraction technique used to detect geometric shapes by transforming points in the image space to curves in the parameter space.
3. **Parameter Space**: The space where each point represents a potential shape in the image

Steps in Edge Linking via Hough Transform

1. Edge Detection

First, apply an edge detection algorithm to identify edges in the image.

2. Hough Transform for Line Detection

Apply the Hough transform to detect lines in the edge-detected image.

3. Edge Linking

Link edges by combining the detected lines to form continuous edges.

Thresholding

Thresholding is one of the simplest and most commonly used techniques in image segmentation. It involves converting a grayscale or color image into a binary image by selecting a threshold value that separates pixels into two categories: those above the threshold and those below it.

Here's how thresholding works:

1. **Grayscale Conversion (if necessary)**: If the input image is not already in grayscale, it is first converted to a single-channel grayscale image. This step simplifies the thresholding process by reducing the image to a single intensity value per pixel.
2. **Threshold Selection**: A threshold value is chosen based on the characteristics of the image and the segmentation task. This value separates the intensity levels into two groups: those above the threshold (foreground) and those below it (background).
3. **Pixel Classification**: Each pixel in the grayscale image is then compared to the threshold value. If the pixel intensity is greater than the threshold, it

is assigned a value of 255 (white or foreground); otherwise, it is assigned a value of 0 (black or background).

4. **Binary Image Formation**: The result is a binary image where pixels are either completely black (0) or completely white (255), depending on whether they are above or below the chosen threshold.

Thresholding can be applied using various techniques:

- **Global Thresholding**: A single threshold value is applied to the entire image.
- **Adaptive Thresholding**: Different threshold values are calculated for different regions of the image, taking into account local variations in intensity.
- **Otsu's Method**: Automatically calculates an optimal threshold value based on the image histogram, aiming to minimize intra-class variance.
- **Multi-level Thresholding**: Divides the intensity range into multiple thresholds, enabling segmentation of images with multiple regions of interest.

Region based segmentation

Region-based segmentation is a technique in image segmentation that divides an image into regions based on predefined criteria. The main goal is to group together pixels that share similar attributes such as intensity, color, texture, or other statistical characteristics, forming contiguous regions. Here's an overview of the key concepts and methods involved in region-based segmentation:

Common Methods

1. Region Growing

- **Seed Selection**: Start with seed points, which are chosen based on some criteria (e.g., manually, random, or using some automatic method).
- **Growth**: Expand the regions by appending neighboring pixels that have similar properties to the seed points.
- **Stopping Criteria**: The growth stops when no more pixels satisfy the homogeneity criteria.

2. Region Splitting and Merging

- **Quadtree Decomposition**:

- **Splitting**: Divide the image into quadrants recursively until each quadrant meets a homogeneity criterion.
- **Merging**: Adjacent regions are merged if their combination meets the homogeneity criterion.

- **Iterative Splitting and Merging**:
  - **Splitting**: Start with the whole image and split it into smaller regions until the homogeneity criterion is met.
  - **Merging**: Merge neighboring regions if the combined region meets the homogeneity criterion.

Morphological processing

Morphological processing is a technique in image analysis and computer vision that involves the application of morphological operations to process and analyze images, particularly for segmentation purposes. These operations are based on the shape or structure of objects within an image and are typically applied to binary or grayscale images. The fundamental morphological operations include erosion, dilation, opening, and closing, which are used to manipulate the structure of objects in the image.

Fundamental Morphological Operations

1. Erosion

- **Effect**: Shrinks objects in the image.
- **Operation**: The pixel is set to 1 only if all the pixels under the structuring element are 1; otherwise, it is set to 0.
- **Usage**: Removes small noise, detaches connected objects.

2. Dilation

- **Effect**: Expands objects in the image.
- **Operation**: The pixel is set to 1 if any pixel under the structuring element is 1.
- **Usage**: Fills small holes, connects adjacent objects.